

Characterize a Declared Fixture

Learn brackets, then verify without circular reasoning

E0 COPIED REPLAY — NO MODULE, ADC, OR COMPARATOR IS POWERED

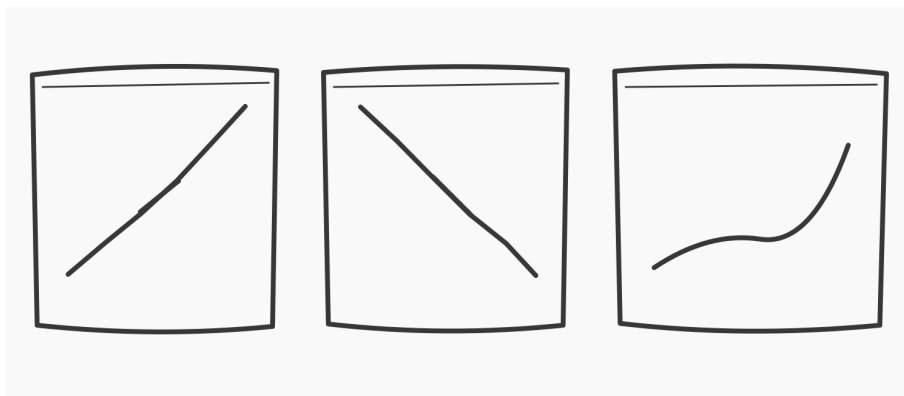
Lesson 071 — component
Prerequisite Lesson 070 descriptor/frame
Input three copied point streams

Time 180–240 minutes
Board compile-only Mega replay
Observation typed evidence cells

Responsibility. `ModuleCharacterizationPolicy` streams one ascending learning leg, one descending learning leg, and one independent verification leg. It publishes sampled transition brackets, conservative intervals, witnesses, and relation evidence. It owns no endpoint, clock, stimulus, power, storage, or display.

1 Ask only what copied evidence can answer

A finite sweep cannot reveal an exact threshold or prove that a physical pin is stuck. It can show that one copied comparator change occurred between two adjacent copied raw codes. Lesson 071 preserves that uncertainty instead of averaging it away.



Alternative text: three graphite notebook lanes run in order: ascending learns one bracket, descending learns another, and verification checks fresh copied points against the already frozen intervals.

Published at E0

copied point order and provenance
 adjacent sampled transition brackets
 bounded conservative intervals
 deterministic disagreement evidence

Still open at E1

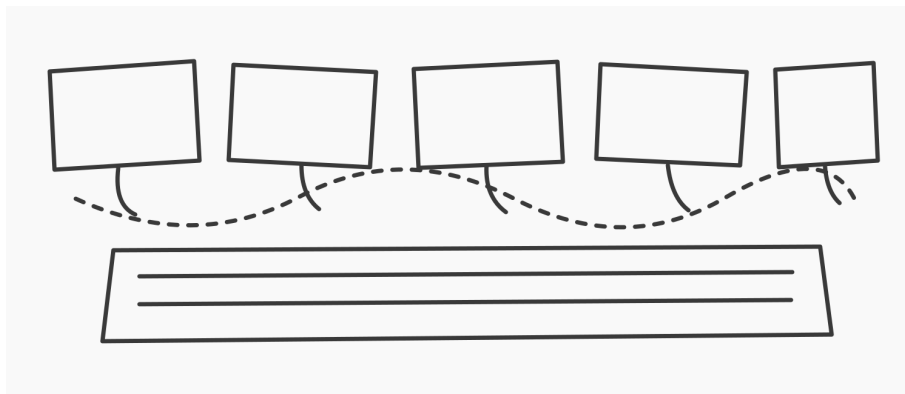
exact board identity and pinout
 electrical threshold or voltage accuracy
 powered acquisition and stimulus
 physical diagnosis and bench acceptance

2 Predict the experiment

Why must verification come after both learning legs? _____. If a transition lies between raw codes 410 and 430, what exact threshold has been proved? _____.

3 Bind every point to one run

ModuleCharacterizationPoint carries nonzero session, run, and leg identities; a contiguous control ordinal; leg and direction; source and configuration identity; and one complete **ModuleThresholdFrame**. Every point must match the copied descriptor, specimen and evidence revisions, source, active leg, and expected ordinal.



Alternative text: a graphite thread stitches descriptor, specimen, source, session, run, leg, ordinal, sequence, and time labels onto each point card before it may enter a sweep.

Each leg accepts exactly **requiredPointsPerLeg**, configured from 2 through 16. Ordinals are independently contiguous 1..N. Source sequences remain contiguous forward across all accepted points. A fieldwise identical duplicate is idempotent; a changed duplicate, gap, regression, identity drift, or wrong ordinal changes nothing and follows the documented failure disposition.

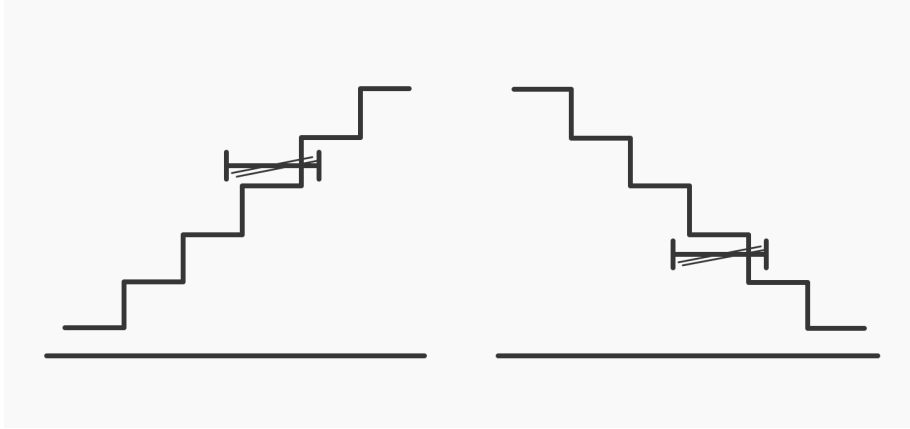
4 Use the frame as readiness authority

The complete Lesson 070 frame is the sole copied warm-up and settling authority. A point does not add another readiness flag. Both channels must be present and current, both producer statuses must be OK, and comparator assertion must agree with explicit active-high or active-low polarity.

Vocabulary discipline. “Increasing” and “decreasing” describe raw ADC-code order only. They do not mean increasing light, heat, sound, distance, hazard, or concentration.

5 Learn one bracket in each direction

Ascending accepts **Increasing**; descending accepts **Decreasing**. On the first canonical comparator change, the immediately adjacent points become the complete **ModuleTransitionBracket**. The policy never sorts, interpolates, reverses, retries, or discards points.



Alternative text: an increasing staircase brackets a comparator change between adjacent samples, while a decreasing staircase brackets a separate change; shaded graphite between samples marks uncertainty rather than an exact crossing.

Observation	Result
one adjacent state change	retain the complete bracket
second reversal	Chatter
toggle at equal raw code	Chatter
wrong raw direction	DirectionViolation
no transition	active/inactive no-observation reason
endpoint-only no-transition leg	AtLowerRail or AtUpperRail

A healthy raw endpoint is ordinary accepted evidence and may participate in a bracket. Rail classification happens only when finalizing an endpoint-only no-transition leg. It never diagnoses an open, short, reversed supply, absent pull, or damaged module.

6 Retain intervals, not invented precision

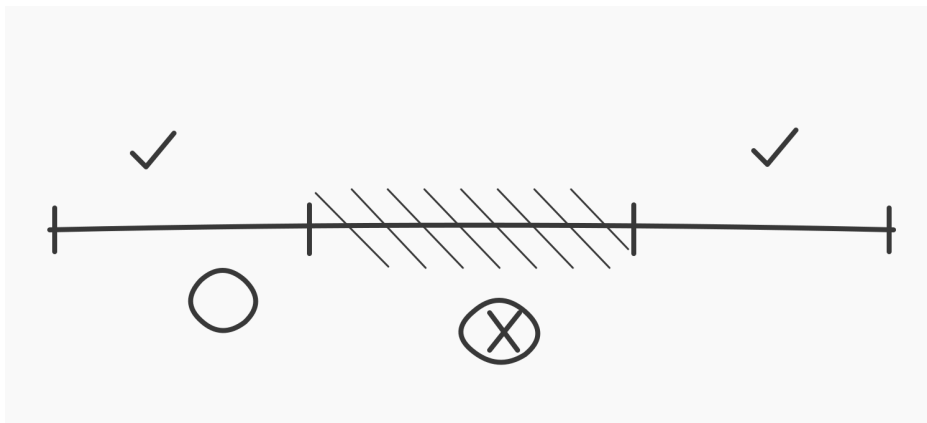
Every learning leg must cover the complete declared raw domain. Ascending begins exactly at the lower bound and ends exactly at the upper bound; descending begins at the upper bound and ends at the lower bound. A missing endpoint records **DirectionViolation**, except when every accepted point is the same exact declared endpoint. That endpoint-only no-transition special case records **AtLowerRail** or **AtUpperRail** before the coverage check, terminalizes the run, cannot advance to another leg, and produces no intervals. Every successful learning leg therefore covers the full domain, preventing outer guaranteed intervals from extrapolating a state beyond observed sweep coverage.

Before intervals exist, the two brackets must have complementary orientation: ascending-before equals descending-after, ascending-after equals descending-before, and those two states differ. Otherwise descending finalization records **TransitionOrientation Mismatch**.

Let **lowState** be the common ascending-before/descending-after state. Let **lowProved** be the lesser raw code of those two points, and **highProved** the greater raw code of ascending-after and descending-before. The policy freezes inclusive **guaranteedInactiveInterval**, **guaranteedActiveInterval**, and **ambiguityInterval** values. The interval from the declared lower bound through **lowProved** proves **lowState**; the interval from **highProved** through the declared upper bound proves the opposite. The closed middle $[\text{lowProved} + 1, \text{highProved} - 1]$ is ambiguity only when its lower endpoint does not exceed its upper endpoint. Arithmetic widens before adding or subtracting. The two outer intervals are assigned active/inactive according to **lowState**. The public contract has no bracket midpoint, exact threshold, or signed hysteresis scalar.

7 Verify without learning from the answer

Verification accepts **Unordered** copied points only after both brackets and intervals are frozen. It never moves either learned bracket.



Alternative text: a pencil number line separates guaranteed inactive and guaranteed active regions with a cross-hatched ambiguity region; matching marks sit in guaranteed regions, an uncertain mark stays ambiguous, and one contradiction is circled.

- A matching state in a guaranteed region contributes **Consistent**.
- A contradictory state in a guaranteed region produces **AnalogComparatorDisagreement** and **Disagrees**.
- A point inside ambiguity contributes **Ambiguous**, never disagreement merely because it is near a sampled transition.
- If every verification point is ambiguous, the relation is **Ambiguous**, not **Consistent**.

Predict the relation for a contradictory point inside ambiguity: _____. Why is that answer conservative? ____.

8 Keep streaming storage bounded

The policy retains no point array. It owns counts, run correlation, adjacent transition evidence, frozen intervals, relation state, and compact witnesses. Exactly four full points remain, inside the two brackets. The **firstWitness** and **lastWitness** compact values are the sole canonical owners of first/last sequence and observation time; evidence has no duplicate top-level first/last fields. Offending diagnostics are compact witnesses too.

Measured item	Actual	Target	Hard	Disposition
no-LTO flash	11,562 B	12,288 B	16,384 B	pass
static SRAM	1,160 B	1,024 B	1,536 B	accepted miss
synchronous stack	339 B	448 B	640 B	pass
policy object	498 B	512 B	768 B	pass
evidence object	375 B	320 B	384 B	accepted miss
phase-local point	57 B	96 B	128 B	pass
residual SRAM	6,565 B	4,096 B	3,072 B	pass

The two target misses are independently accepted because the complete attributable evidence shape must retain both brackets, the full descriptor, three intervals, correlation, and compact witnesses, while the canonical sketch deliberately exposes one stable evidence object for non-Serial inspection. Both remain below their hard limits. The no-LTO measurement is bound to resource fingerprint **e1cfc001cc95937e25337eb53a7a4c8b3602d6a046b23b0a8b94fe7a342d562b**.

9 Follow the lifecycle in dependency order

Construction is inert. The canonical flow is:

1. `initialize(now)`, then `beginSession(now, sessionId, runId)`;
2. begin, observe exactly N points, and finalize `Ascending`;
3. repeat for `Descending`;
4. repeat for `Verification`; and
5. copy immutable terminal evidence with `evidence(output)`.

Leg IDs are nonzero, distinct, forward modular, and ordered ascending, descending, verification. Initialize and reset advance a nonzero lifecycle generation without wrap. Every successful lifecycle call retains its supplied `now`; later session, leg, finalization, reset, and shutdown calls require forward-or-equal modular time. Backward and exact-half-range ambiguous times reject atomically. Shutdown is permanent for that object: `initialize()` after shutdown returns `NotInitialized`; only a newly constructed policy begins another object lifecycle. An incomplete finalize or a point beyond N rejects atomically; capacity is not a fabricated domain observation.

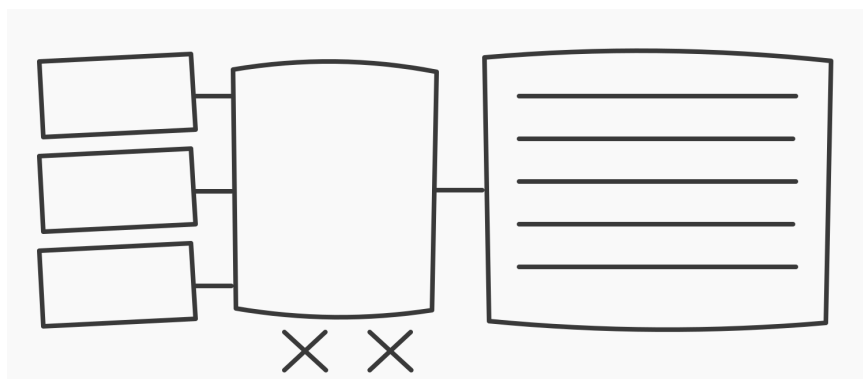
10 Apply failure precedence before interpretation

1. malformed configuration or encoding;
2. lifecycle, identity, correlation, or ordinal mismatch;
3. attributable producer failure;
4. attributable time or sequence discontinuity;
5. attributable `WarmupUnsatisfied` or `SettlingUnsatisfied` copied readiness evidence;
6. direction, transition, chatter, interval, or relation outcome; and
7. complete verification evidence.

Analog producer failure dominates comparator interpretation. Comparator producer failure dominates chatter and relation analysis. Ordering failure dominates an apparent crossing. Every failed API call leaves policy evidence and caller output unchanged.

11 Inspect one deterministic replay

The Mega sketch acquires copied fixture arrays, configures one `ModuleCharacterizationPolicy`, starts no hardware, streams all three legs, then writes named volatile status, bracket, interval, relation, witness, count, and provenance cells.



Alternative text: three copied fixture-card stacks feed a bounded policy notebook; the same bracket, interval, witness, and relation cells appear on two replay ledgers while crossed-out wires and power remain outside.

Predict the two brackets and three intervals before compilation. Observe the named cells with a debugger, simulator, or disassembly; Serial is optional. Repeat the replay and compare all cells. Agreement proves deterministic copied behavior, not electrical truth.

12 Test the edges, collisions, and non-mutation

The current deterministic host matrix exercises a reduced raw domain at its lower, middle, and upper boundaries; complete ascending, descending, and verification flow; clean complementary brackets; inclusive guaranteed and ambiguity boundaries; consistent, ambiguous, and disagreeing verification; readiness failures; endpoint-only terminal results; identity, ordering, capacity, and lifecycle rejection; and output non-mutation. It does not claim every possible bracket ordering or every value in the full 16-bit raw domain. Additional exhaustive families remain explicit promotion work when not represented by those boundary partitions.

13 Troubleshoot without diagnosing hardware

Symptom	Inspect	Honest interpretation
point rejected	IDs, ordinal, descriptor, source	correlation failure
leg rejected	point count and direction	incomplete or wrong ordered leg
Chatter	adjacent comparator states/raw codes	copied reversal evidence
no transition	observed canonical state	finite sweep saw no crossing
AtLowerRail	all accepted endpoint values	endpoint-only copied leg
TransitionOrientationMismatch	both learned brackets	states are not complementary
Ambiguous	frozen interval and point	learned evidence cannot decide
Disagrees	guaranteed region and witness	copied relation contradiction

14 Exercises

1. Draw brackets for a rising change from 390 to 420 and a falling change from 370 to 400. Mark only intervals justified by both.
2. Explain why a healthy value equal to the raw-domain maximum is not automatically **AtUpperRail**.
3. Create a verification set containing only ambiguity points and predict the final relation.
4. Explain why exactly four full bracket points are retained but 48 input points are not.
5. Name the E1 evidence required before a copied disagreement can support any electrical diagnosis.

15 No circuit and no powered stimulus at E0

There is no wiring diagram or formal schematic because Lesson 071 has no circuit. Never connect or power an unidentified module. Exact E1 work must qualify the specimen and declarations, supply and logic limits, connector, pull/output topology, ADC reference, source impedance, inactive default, harmless family-specific stimulus, observation points, rollback, power removal, and measured acceptance before powered acquisition exists.

No gas exposure, heater, flame, combustion, medical or physiological claim, body contact, capacitive-touch assumption, immersion, mains, high voltage, laser, ignition, pyrotechnics, alarm authority, or unattended stimulus is admitted.

16 Open acceptance record

Host evidence	Record strict tests, sanitizers, standalone header, the reduced-domain three-leg boundary matrix, atomicity, and zero endpoint/clock/power calls; do not label it exhaustive.
Mega evidence	Record compile-only replay flash/static SRAM and named-cell inspection; this is not a powered-module result.
Memory evidence	Fingerprint-bound no-LTO result: 11,562 B flash; 1,160 B static SRAM; 339 B stack; 498 B policy; 375 B evidence; 57 B point; 6,565 B residual SRAM. Static and evidence target misses are independently accepted below hard limits.
Document evidence	Two-pass build, metadata, embedded fonts, text extraction, monochrome, visual classification, links, 200% zoom, and rendered-pencil review must pass.
Physical evidence	Open E1: exact identity, safe acquisition and stimulus, declaration truth, timing, observations, rollback, and bench acceptance.

Acceptance item	Record
Commit and toolchain	
Host / sanitizer / matrix	
Mega flash / static / stack	
Policy / evidence / point sizes	
Named replay cells	
PDF / HTML human review	
Exact E1 owner and next action	

Stop condition. Call this bounded characterization of copied evidence. Do not call it an exact threshold, calibrated sensor, electrical fault diagnosis, qualified module, powered experiment, durable record, or physical bench result.

Next: Lesson 072 — Compose one inert module-characterization bench.